Course Schedule: Mechanics and Relativity WBPH001-10 2026:1a

Based on David Morin's *Introduction to Classical Mechanics*

Overview

- **Lectures:** Mondays 09:00–11:00, Wednesdays 13:00–15:00
- **First Lecture:** Wednesday, September 2, 2026
- **Midterm Exam:** Friday, October 2, 18:30–20:30 (Lectures 1–8)
- **Final Exam:** Monday, October 26, 18:15–20:15 (Comprehensive)
- **Textbook:** David Morin, *Introduction to Classical Mechanics*

Lecture Schedule

Lec.	Date	Time	Topic	Reading
1	Wed Sep 2	13–15	Course overview; introduction and historical context	Intro / supplementary
2	Mon Sep 7	09–11	Foundations of special relativity: events, frames, and postulates	Ch. 11.1–11.3
3	Wed Sep 9	13–15	Lorentz transformations and basic consequences	Ch. 11.4–11.5
4	Mon Sep 14	09–11	The invariant interval, proper time, spacetime diagrams	Ch. 11.6–11.7
5	Wed Sep 16	13–15	The Doppler effect, rapidity, and the pole-barn paradox	Ch. 11.8–11.10
6	Mon Sep 21	09–11	Chapter 11 recap: spacetime diagrams and two-observer problems	Ch. 11
7	Wed Sep 23	13–15	Dimensional analysis and natural units	Supplementary
8	Mon Sep 28	09–11	Fermi problems and estimation techniques	Supplementary
9	Wed Sep 30	13–15	Newtonian dynamics review: forces, energy, and momentum	Ch. 2, 3, 5
–	Fri Oct 2	18:30–20:30	**Midterm Exam (Lectures 1–8)**	
10	Mon Oct 5	09–11	Relativistic energy, momentum, and transformation rules	Ch. 12.1–12.2
11	Wed Oct 7	13–15	Collisions, decays, and particle-physics units	Ch. 12.3–12.4
12	Mon Oct 12	09–11	Relativistic force and rocket motion	Ch. 12.5–12.6
13	Wed Oct 14	13–15	4-vectors: definitions, examples, and properties	Ch. 13.1–13.3
14	Mon Oct 19	09–11	4-vectors continued: energy, momentum, force, and physical laws	Ch. 13.4–13.6
15	Wed Oct 21	13–15	Final comments: equivalence principle and accelerated frames	Ch. 14
–	Mon Oct 26	18:15–20:15	**Final Exam (Comprehensive)**	